

Sanapathi Sai Kumar

B.Tech CSE (AI & ML) | Full-Stack Developer | Software Engineering

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SUMMARY

Final-year B.Tech CSE (AI & ML) student and Associate Software Engineer with production experience building healthcare applications. Backend-leaning full-stack developer experienced in REST APIs, JWT authentication, role-based access control and real-time systems using Node.js, Express.js, MongoDB and React. Meta PyTorch OpenEnv Hackathon finalist — top 3% of 31,000+ teams.

EDUCATION

B.Tech, Computer Science and Engineering (AI & ML)

2023 – 2027 (Expected)

Dhanalakshmi Srinivasan University

CGPA: 7.0/10, No Backlogs | Intermediate: 83% | Class X: 91%

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, Java, SQL

Backend: Node.js, Express.js, REST APIs, JWT Authentication, Role-Based Access Control, FastAPI

Frontend: React.js, Next.js, HTML5, CSS3, Tailwind CSS

Databases & Real-time: MongoDB, Firebase, Socket.IO, WebSockets

Tools: Git, GitHub, Docker, Postman, Figma

AI/ML (project-based): PyTorch, FAISS, sentence-transformers, NLP

EXPERIENCE

Associate Software Engineer, Lystra Pharma Private Limited | Remote

Apr 2026 — Present

HoMedi — home healthcare booking and service management platform (live in production)

- Develop and maintain REST APIs and backend services (Node.js, Express.js, MongoDB) for the booking flow, service catalog and provider-management modules of a live production healthcare platform
- Implement JWT authentication and role-based authorization separating patient, provider and administrator access across all protected routes
- Build responsive, mobile-first React dashboards for all three user roles, integrate them with backend APIs, and iterate on features with the product and design team

UX and UI Designer Intern, Zidio Development | Remote

Jul 2025 — Oct 2025

- Designed wireframes, user flows and high-fidelity interfaces in Figma and Adobe XD, and ran usability testing

Frontend Developer Intern, Cognifyz Technologies | Remote

Mar 2025 — Apr 2025

- Built responsive React.js interfaces, integrated REST APIs, and optimized asset loading and rendering to improve frontend page-load performance

PROJECTS

VAYU-DRISHTI, Climate Digital Twin OS for India | Personal Project | GitHub

Next.js, Express.js, MongoDB, FastAPI, PyTorch

- Architected a three-tier system — Next.js dashboard, Express.js/MongoDB API gateway and FastAPI ML service — with all model requests proxied through the gateway and every scenario persisted, keeping the ML service isolated from clients
- Served ConvLSTM deep-learning forecasts (1-, 7- and 30-day horizons over a 31×31 grid) on an interactive MapLibre GL map with temporal scrubbing, live telemetry over Socket.IO with REST polling fallback, and a drought/flood what-if simulator

Smart Mentor, Real-Time Adaptive Learning Platform | SIH 2025, Team Lead | GitHub

Next.js, FastAPI, WebSockets, FAISS

- Led Team INVOTEX and built the real-time layer end-to-end: FastAPI WebSocket services streaming live learning signals to Next.js clients on each channel
- Implemented Bayesian Knowledge Tracing — a probabilistic model that estimates in real time which concepts a student has mastered — and semantic doubt retrieval over a FAISS vector index using sentence-transformers embeddings

OpenEnv Invotex, RL Environment for LLM-Agent Benchmarking | Meta PyTorch Hackathon | GitHub

Python, Docker

- Built OpenEnv-compliant reinforcement-learning environment APIs across three enterprise domains (Email Triage, Customer Support, Traffic Control) with a deterministic, Docker-containerized evaluation harness so agent runs stay repeatable and comparable
- Trained agents against injected failure — live schema drift, 403s and permission errors — cutting training loss 3.8 to 0.14 in 60 steps, with LLaMA-3 8B scoring 0.65+ where Qwen-72B scored 0.00

HACKATHONS & ACHIEVEMENTS

- Smart India Hackathon 2025:** Team Lead. Built Smart Mentor.
- Meta PyTorch OpenEnv Hackathon:** Grand Finalist, top 3% of 31,000+ teams globally. Built OpenEnv Invotex.
- Cardano Asia Hackathon — IBW 2025:** Finalist.
- Bengaluru AI Hack Day:** Solo Builder. Built ContentOps AI.

CERTIFICATIONS

MEAN Stack Development — GUVI (HCL) | Fundamentals of Machine Learning and Artificial Intelligence — AWS Training & Certification | Generative AI & AI Foundations — Microsoft Learn (3 courses)